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Protective effects of *Fumaria parviflora* L. on lead-induced testicular toxicity in male rats

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Summary

In recent years, the clinical importance of herbal drugs has received considerable attention in reducing free radical-induced tissue injury. Oxidative stress has been proposed as a possible mechanism involved in lead toxicity that causes reproductive system failure in both human and animals. *Fumaria parviflora* L., a traditional herb, has been used to cure various ailments in Persian folk medicine. This study was carried out to investigate whether ethanolic extract of *F. parviflora* leaves could protect the male rats against lead-induced testicular oxidative stress. Adult Wistar rats were treated with 0.1% lead acetate in drinking water with or without 200 mg kg day^{-1} *F. parviflora* extract via gavage for 70 days. Lead acetate treatment resulted in significant reduction in testis weight, seminiferous tubules diameter, epididymal sperm count, serum testosterone level, testicular content of superoxide dismutase (SOD) and glutathione peroxidase (GPx). Moreover, significant elevation was observed in content of malondialdehyde (MDA) in lead-treated rats. However, co-administration of *F. parviflora* extract showed a significant increase in selected reproductive parameters in lead-treated rats. The results indicated that ethanolic extract of *F. parviflora* leaves has a potential to restore the suppressed reproduction associated with lead exposure and prevented lead-induced testicular toxicity in male Wistar rats.

Introduction

Over the past few decades, several evidences have shown that deterioration in male reproductive health, particularly a global decline in semen quality, leads to increasing in prevalence of infertility. Infertility affects approximately 15% of couples of reproductive age and with nearly half of these cases resulting from male factor. Human, animal and *in vitro* studies suggest that environmental and occupational exposures to heavy metals may have adverse impacts on male reproductive health (De Rosa *et al.*, 2003; Hernández-Ochoa *et al.*, 2005; Meeker *et al.*, 2008), even at relatively low-level exposures (Telisman *et al.*, 2007). Heavy metals could adversely affect the male reproductive system either by causing hypothalamic–pituitary axis disruption or by affecting spermatogenesis directly, resulting in impaired semen quality (Wyrobek *et al.*, 1997). Lead is known as reproductive toxicant for humans and other mammals, which released into the environment by several routes, absorbed via respiratory, gastrointestinal tracts and through the skin and accumulated in the body over lifetime. To date, animal studies have shown that lead can induce abnormalities in adult male reproductive function (Sokol *et al.*, 1985; Wadi & Ahmad, 1999; Gorbel *et al.*, 2002; Batra *et al.*, 2004; Graca *et al.*, 2004). A number of studies have reported a significant negative correlation between blood and seminal lead concentrations and semen quality among both occupationally exposed and unexposed men (De Rosa *et al.*, 2003; Jurasović *et al.*, 2004; Eibensteiner *et al.*, 2005). Reproductive toxicity of low-level lead exposure in men with no occupational exposure to metals was reported by Telisman *et al.* (2000). It was demonstrated that lead toxicity to be related to increasing the level of lipid peroxidation (Upasani *et al.*, 2001), generation of reactive oxygen species (Marchlewicz *et al.*, 2004; Ni *et al.*, 2004) and inhibition of the antioxidant enzyme activities (Bolin *et al.*, 2006). Thus, it has been supposed that the treatment with antioxidant compounds may be to counteract the adverse effects of lead on different body systems (Mishra & Acharya, 2004; Shalan *et al.*, 2005).

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In this respect, supplementation with certain plants is considered to reduce toxic effects of various toxicants including heavy metals (Nandi *et al.*, 1997). *Fumaria parviflora* Lam. (Fumariaceae) is a herbaceous plant that grows in wide variety parts of Iran, Indo-Pakistan subcontinent and Turkey. It is known as Shahtareh in Iran, and aerial parts of *F. parviflora* have been used traditionally in Iranian folk medicine against liver and bile duct disorders and dermatological diseases such as scabies, eczema and acne and as antiscorbite, antibronchite, diuretic, expectorant, antipyretic, diaphoretic, appetiser and antineoplastic (Amin, 1991; Zargari, 1996). It has also been supposed traditionally that *F. parviflora* has male fertility-promoting properties (Amin, 1991). Studies have been reported that *F. parviflora* (Jamshidzadeh & Niknahad, 2006) and *F. vaillantii* (Mortazavi *et al.*, 2007), another species of genus *Fumaria,* have ability to counteract CCl_4-induced hepatotoxicity with its antioxidative properties. Protective effect of *F. parviflora* against paracetamol-induced hepatotoxicity has also been demonstrated (Gilani *et al.*, 1996). In addition, antinociceptive effect of *F. parviflora* percolated extract in mice and rats (Heidari *et al.*, 2004) and its beneficial effect against hand eczema have been shown (Jowkar *et al.*, 2011). Venkateswara Rao *et al.* (2007) found that ethanolic extract of *Fumaria indica*, another species of genus *Fumaria*, has anti-inflammatory and antinociceptive activities. Also, it was demonstrated that *F. parviflora* has favourable effect on spermatogenesis in male rats (Heydari Nasrabadi *et al.*, 2012). Phytochemical analyses of some plants of genus *Fumaria*, including *F. parviflora,* have indicated presence of isoquinoline alkaloids namely protropine, cryptopine, sinactine, stylopine, bicuculline, adlumine, parfumine, fumariline, fumarophycine, fumaritine, dihydrofumariline, perfumidine and dihydrosanguirine (Suau *et al.*, 2002). Meanwhile, studies of Soušek *et al.* (1999) showed that phenolic extracts from *Fumaria* were 2–3-fold more effective as scavengers of 1,1-diphenyl–2-picrylhydrazyl radical than the alkaloid extracts. Therefore, this study was undertaken to investigate effect of *F. parviflora* ethanolic extract on the oxidative stress and reproductive toxicity induced by lead acetate in Wistar male rats.

Materials and methods

Plant materials

The leaves of *Fumaria parviflora* Lam. were collected from Kazeroun city in Fars province of Iran in March 2012. It was recognised and authenticated in Botanical Systematic Laboratory, Department of Biology, Shahid Chamran University of Ahvaz, Iran.

Preparation of ethanolic extract of Fumaria parviflora Lam

Fresh leaves of *F. parviflora* were dried at 40 °C for 48 h. The dried leaves (200 g) were powdered in an electrical grinder and extracted successively with ethanol (600 ml) in a Soxhlet extractor for 48 h at 60 °C. After extraction, the mixture was filtered and the solvent was evaporated to dryness at 50–55 °C using a rotary evaporator and the extract left behind (yield was 5.6 g kg^{-1}) was stored at 4 °C. It was dissolved in distilled water whenever needed for experiments.

Chemicals

Lead acetate was purchased from Sigma-Aldrich, (St. Louis, MO, USA). Glutathione peroxidase (GPx) and superoxide dismutase (SOD) kits were purchased from Randox Labs Ltd. (Ardmore, UK). Testosterone, luteinising hormone and follicle-stimulating hormone assay kits were purchased from Monobinde Inc., CA, USA. All other chemicals used in study were purchased from Merck, Darmstadt, Germany.

Animals

Forty healthy adult male Wistar rats (8 weeks old, 180–200 g body weight) were obtained from Research Center and Experimental Animal House of Ahwaz Jundishapour University of Medical Sciences. The animals were housed under standard laboratory conditions (temperature 23 ± 2 °C, relative humidity of 50 ± 5%, 12-h/12-h light/dark cycle) and were given food and water *ad libitum*. The Animal Ethics Committee of Shahid Chamran University of Ahvaz has approved the experimental protocol.

Experimental design

After 1 week of acclimatisation, animals were divided randomly into four equal groups (n = 10) as follows:

Group 1 (control) was administered 1 ml distilled water via gavage once daily for 70 days.

Group 2 was administered 1 ml distilled water via gavage once daily and received distilled water containing 0.1% lead acetate *ad libitum* for 70 days. Male rats were exposed to lead acetate for 70 days to evaluate its effect through a complete spermatogenic cycle (Hayes, 1989).

Group 3 was administered with ethanolic extract of *F. parviflora* at dose 200 mg kg^{-1} body weight via gavage once daily.

Group 4 was administered with ethanolic extract of *F. parviflora* at dose 200 mg kg^{-1} body weight via gavage

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once daily and received distilled water containing 0.1% lead acetate *ad libitum* for 70 days.

The doses for lead acetate were selected on the basis of previously published studies (Ronis *et al.*, 1996; El-Sayed & El-Neweshy, 2010).

The body weight of rats was recorded at the termination of the experiment. After the administration of the last treatment, the animals were fasted overnight, and then, on the next day, they were sacrificed under light ether anaesthesia. Blood samples were collected by heart puncture for determination of lead and testosterone levels. The reproductive organs, testes, epididymis, seminal vesicles and ventral prostate were removed and weighed. Right testes immediately fixed into Bouin's solution, and left testes were minced and homogenised for various biochemical assays.

Measurement of blood lead level

Blood samples were collected from the heart for estimation of blood lead level by graphite furnace atomic absorption spectrometry (GFAAS) in the Kimiay-e-Nab Analysis laboratory, Karaj, Iran.

Morphometrical analysis

Tissue samples from right testes were excised and processed for paraffin embedding sections. Serial sections with 5 μm thickness were stained with haematoxylin and eosin and used for morphometrical studies at light microscopic level. For measuring of seminiferous tubule diameter and germinal epithelium height, 90 round or nearly round cross sections of seminiferous tubules were randomly chosen in each rat. Then, using an ocular micrometer of light microscopy (Olympus BH, Tokyo, Japan), at a magnification of ×40, two perpendicular diameters of each seminiferous tubule cross section were measured and the mean of these was calculated. Also, germinal epithelium height in four equidistance of each seminiferous tubule cross section was measured and the mean of these was calculated (Franca & Godinho, 2003).

Sperm analysis

Epididymis was separated carefully from testis and divided into three segments: head, body and tail. The epididymal tail was trimmed with scissors and placed in Petri dishes containing 1.0 ml of 0.1 M phosphate buffer of pH 7.4. The dishes gently swirled for homogeneity and allowed sperm diffusion in the solution for 10 min under 37 °C for dispersion of sperm cells. Sperm samples were assessed for motility, number and gross morphology of spermatozoa without the investigator knowing which samples were from which group. For analysis of sperm motility, an aliquot of 10 μl was placed in a hemocytometer chamber (Paul Marienfeld GmbH, Lauda-Königshofen, Germany) and analysed under a light microscope. One hundred spermatozoa were evaluated per animal and classified into motile and immotile (Seed *et al.*, 1996). For evaluation of sperm concentration, an hour after the sperm diffusion in the solution, a 10-μl aliquot of the epididymal sperm suspension was transferred to each counting chamber of the haemocytometer and allowed to stand for 5 min. Then, sperm heads were counted by a light microscope at ×40 magnification and expressed as million ml^{-1} of suspension (Seed *et al.*, 1996). The sperm morphology was also determined using eosin staining method. For this purpose, 10 μl of 1% eosin Y was added to a test tube containing 40 μl of sperm suspension and was mixed by gentle agitation. Then, spermatozoa were incubated at room temperature for 45–60 min for staining and then resuspended with a Pasteur pipette. Two hundred spermatozoa per animal were examined microscopically at ×40–100 magnifications, and the number of morphologically abnormal spermatozoa was recorded to give the per cent of abnormal spermatozoa.

Biochemical assays

The left testes were removed and dissected free from the surrounding fat and connective tissue. Each testis was longitudinally sectioned, kept at −80 °C and subsequently homogenised (10% w/v) in ice-cold 0.1 M sodium phosphate buffer. The testicular homogenates were centrifuged at 13000 ***g*** for 15–20 min at 4 °C twice to get enzyme fraction. The resulting supernatant was used for determination of antioxidant enzyme activity and levels of lipid peroxidation. Glutathione peroxidase levels and superoxide dismutase activity were analysed using colorimetric assay kits according to the recommendations of the manufacturer (Randox Labs Ltd). The amount of produced malondialdehyde (MDA) was used as an index of lipid peroxidation. Malondialdehyde levels in the testis were assayed using the thiobarbituric acid test as described by Ohkawa *et al.* (1979). The absorbance was measured at 532 nm. The levels of lipid peroxidation were expressed as nmol of malondialdehyde per mg of protein.

Serum hormonal assay

Blood samples were left for 60 min to clot and then centrifuged for 10 min at 2430 × ***g***. The obtained clear sera were stored at −80 °C until testosterone, luteinising hormone and follicle-stimulating hormone levels were measured by radioimmunoassay using commercial kit

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(AccuBind ELISA kits, Monobinde Inc., CA, USA). Analyses were carried out according to the kit manufacturer's instructions.

Statistical analysis

The data were analysed using the Statistical Package for Social Science program version 10 (SPSS 10; SPSS Inc., Chicago, IL, USA). Statistical analysis was performed using analysis of variance (ANOVA) followed by Tukey's test. Data are expressed as the mean ± SD, and the level of significance was set at $P < 0.05$.

Results

Body and reproductive organs weight

The treatment of rats with *F. parviflora* caused no change on the body weight of the animals in all the experimental groups. However, the weights of testis, epididymis, seminal vesicle and ventral prostate of the lead acetate-treated rats decreased significantly ($P < 0.05$) as compared to control group. Treatment with *F. parviflora* extract alone led to a significant ($P < 0.05$) increase in the testis and epididymis weight as compared to control rats. Significant increase ($P < 0.05$) in the weights of testis, epididymis, seminal vesicle and ventral prostate was observed in rats co-administrated with *F. parviflora* extract as compared to lead acetate-treated rats (Table 1).

Morphometrical analysis

Histological analysis of the testis revealed that lead acetate caused degenerative changes in the seminiferous tubules epithelium with loss of spermatogenesis, decrease in germinal epithelium height and increase in tubular lumen in male rats. The seminiferous tubules epithelium contained several vacuoles scattered among the spermatogenic cells, and the lumen of some seminiferous tubules did not possess spermatozoa as a result of maturation arrest. Statistically significant ($P < 0.05$) reductions in the seminiferous tubules diameter and germinal epithelium height were observed in rats exposed to lead acetate as compared to control group. In extract-treated alone rats, significant ($P < 0.05$) increase in the seminiferous tubules diameter and germinal epithelium height was observed as compared to control group. Seminiferous tubules diameter and germinal epithelium height were significantly higher ($P < 0.05$) in lead acetate-treated rats co-administrated with *F. parviflora* extract as compared to lead acetate-treated rats (Table 2).

Sperm analysis

Lead acetate administration to rats after 70 days led to a significant ($P < 0.05$) decrease in sperm concentration and per cent of motile spermatozoa. Also, the per cent of spermatozoa with abnormal morphology was statistically ($P < 0.05$) higher in lead acetate-treated rats as compared to control group. Significant ($P < 0.05$) increase in the sperm concentration and decrease in the per cent of spermatozoa with abnormal morphology were observed in extract-treated alone rats as compared to control group (Fig. 1). The sperm parameters improved significantly ($P < 0.05$) in rats co-administrated with *F. parviflora* extract as compared to lead acetate-treated rats (Table 2).

Tissue biochemistry

In rats treated with lead acetate, the testicular MDA levels showed significant ($P < 0.05$) increase when compared with control group. Moreover, the activity levels of SOD and GPx decreased significantly ($P < 0.05$) in the testis of rats exposed to lead acetate as compared to control group. Administration of *F. parviflora* extract alone to male rats significantly ($P < 0.05$) reduced the lipid peroxidation and increased the activity levels of SOD and GPx in the testis as compared to control group. Co-administration of *F. parviflora* extract significantly ($P < 0.05$) enhanced the MDA levels and reduced the activity levels of SOD and GPx in the testis of male rats (Table 3).

Table 1 Effects of *Fumaria parviflora* extract on body and reproductive organs weight in lead acetate-treated male Wistar rats

Parameters	Control	Lead	*F. parviflora*	Lead + *F. parviflora*
Body (g)	322.2 ± 3.1	318.1 ± 6.7	325.3 ± 3.1	320.2 ± 3.1[b]
Testis (mg)	1353.5 ± 15.1	1050.3 ± 14.0[a]	1446.2 ± 15.4[a]	1338.7 ± 14.2[b]
Epididymis (mg)	464.1 ± 6.7	347.6 ± 7.3[a]	552.3 ± 4.2[a]	449.2 ± 5.6[b]
Seminal vesicle (mg)	470.4 ± 5.2	339.8 ± 11.2[a]	468.4 ± 5.4	454.9 ± 4.3[b]
Ventral prostate (mg)	195.7 ± 4.1	148.2 ± 3.1[a]	210.8 ± 3.8	182.2 ± 5.5[b]

Data are presented as mean ± SD.
[a]Differ significantly ($P < 0.05$) with respect to control rats.
[b]Differ significantly ($P < 0.05$) with respect to rats treated with lead acetate.

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Table 2 Effects of *Fumaria parviflora* extract on seminiferous tubule and sperm parameters in lead acetate-treated male Wistar rats

Parameters	Control	Lead	*F. parviflora*	Lead + *F. parviflora*
STD (μm)	254.3 ± 2.1	226.4 ± 3.6[a]	267.8 ± 2.4[a]	250.3 ± 3.2[b]
GEH (μm)	66.8 ± 1.6	49.2 ± 1.5[a]	75.3 ± 1.4[a]	63.2 ± 1.3[b]
Sperm concentration (million ml^{-1})	64.31 ± 2.2	43.56 ± 2.3[a]	76.41 ± 3.4[a]	61.33 ± 3.2[b]
Motile spermatozoa (%)	84.34 ± 3.2	56.54 ± 2.2[a]	86.67 ± 2.7	79.23 ± 3.1[b]
Abnormal spermatozoa (%)	17.9 ± 1.23	31.2 ± 2.1[a]	10.4 ± 1.04[a]	22.5 ± 2.12[b]

Data are presented as mean ± SD.
STD, seminiferous tubule diameter; GEH, germinal epithelium height.
[a]Differ significantly ($P < 0.05$) with respect to control rats.
[b]Differ significantly ($P < 0.05$) with respect to rats treated with lead acetate.

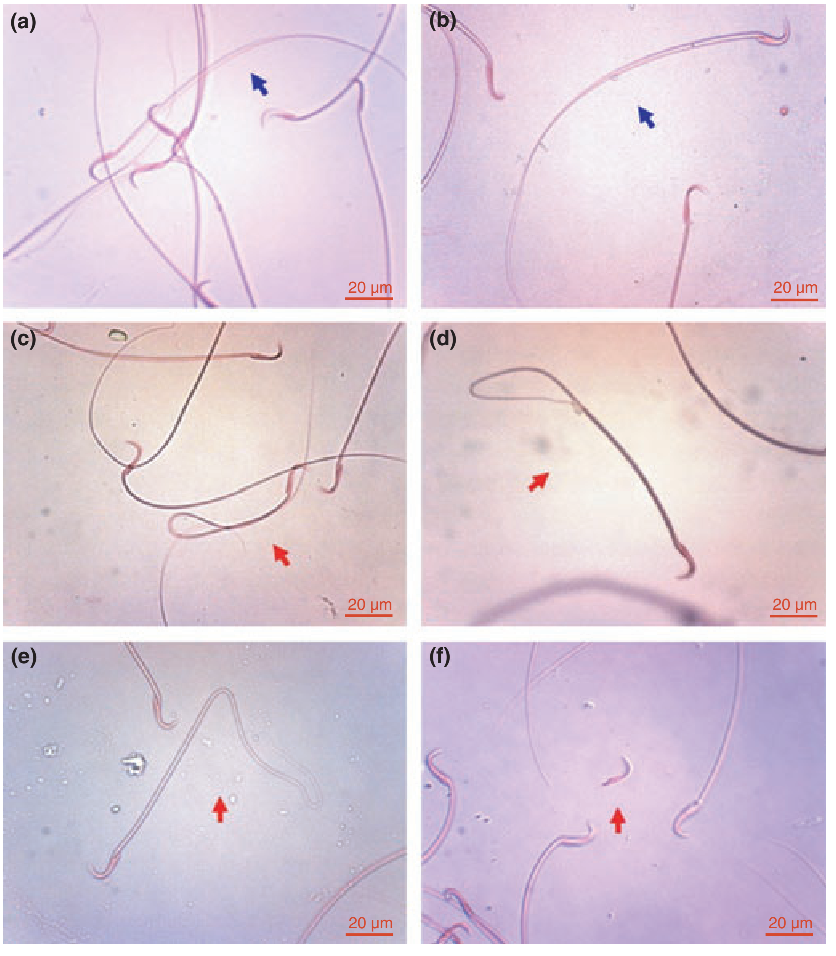

Fig. 1 Spermatozoa in adult male rats with normal (a and b, blue arrows) and abnormal (c: bent midpiece or tail, d and e: bent tail and f: detached head, red arrows) morphology (scale bar: 20 μm, eosin staining).

Serum hormonal assay

Significant ($P < 0.05$) decline in the serum testosterone, FSH and LH levels was observed in lead acetate-treated rats as compared to control group. Testosterone levels increased significantly ($P < 0.05$) in extract-treated alone rats as compared to control group. In rats co-administrated with *F. parviflora* extract, the levels of testosterone, FSH and LH were significantly ($P < 0.05$) higher as compared to lead acetate-treated group (Table 4).

Discussion

During recent decades, concerns have been raised regarding the increases in human infertility that might stem from exposure to environmental and occupational toxicants. Exposure to lead in men, especially in professional workers, results in infertility and sterility (Cullen *et al.*, 1984). Lead has been shown to have adverse effects on male reproductive function directly by changes in

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Table 3 Effects of *Fumaria parviflora* extract on lipid peroxidation and antioxidant enzymes levels in lead acetate-treated male Wistar rats

Parameters	Control	Lead	*F. parviflora*	Lead + *F. parviflora*
MDA (μmol mg^{-1} protein)	8.23 ± 0.52	17.12 ± 0.90[a]	3.14 ± 0.15[a]	10.33 ± 0.61[b]
GPx (nmol mg^{-1} protein)	2.42 ± 0.37	0.81 ± 0.15[a]	5.26 ± 0.22[a]	2.27 ± 0.14[b]
SOD (U mg^{-1} protein)	12.12 ± 0.45	5.18 ± 0.40[a]	25.85 ± 1.11[a]	11.23 ± 0.74[b]

Data are presented as mean ± SD.
MDA, malondialdehyde; GPx, glutathione peroxidase; SOD, superoxide dismutase.
[a]Differ significantly ($P < 0.05$) with respect to control rats.
[b]Differ significantly ($P < 0.05$) with respect to rats treated with lead acetate.

Table 4 Effects of *Fumaria parviflora* extract on serum testosterone, LH and FSH levels in lead acetate-treated male Wistar rats

Parameters	Control	Lead	*F. parviflora*	Lead + *F. parviflora*
Testosterone (ng dl^{-1})	5.66 ± 0.41	1.68 ± 0.30[a]	7.39 ± 0.74[a]	4.67 ± 0.51[b]
LH (ng dl^{-1} or U l^{-1})	0.95 ± 0.04	0.37 ± 0.02[a]	0.88 ± 0.03	0.86 ± 0.04[b]
FSH (ng dl^{-1} or U l^{-1})	3.21 ± 0.05	1.47 ± 0.06[a]	3.08 ± 0.03	2.87 ± 0.04[b]

Data are presented as mean ± SD.
[a]Differ significantly ($P < 0.05$) with respect to control rats.
[b]Differ significantly ($P < 0.05$) with respect to rats treated with lead acetate.

spermatogenesis and sperm function (Rubio *et al.*, 2006) or indirectly by affecting the hypothalamic–pituitary–testicular axis (Sokol *et al.*, 1985). The present study was designed to assess beneficial role of *F. parviflora* leaves extract on lead-induced testicular oxidative stress in Wistar male rats. Male fertility-enhancing properties of *F. parviflora* have been mentioned in Iranian folk medicine (Amin, 1991). The potential role of oxidative stress damage, which is associated with lead exposure (Hermes-Lima *et al.*, 1991; Bechara *et al.*, 1993), suggests that antioxidants may enhance the efficacy of treatment designed to attenuate lead-induced toxicity. In the present study, male rats were exposed to low levels of lead acetate for 70 days to evaluate its effect through a complete spermatogenic cycle (Hayes, 1989). Our findings revealed that co-administration of *F. parviflora* extract prevents lead-induced testicular toxicity and improves reproductive parameters in lead-treated rats. According to the results obtained from this study, exposure to lead acetate disrupts male reproductive function and has adverse effects on reproductive organs weight, testicular structure, epididymal sperm parameters including sperm concentration, motility and morphology and serum testosterone levels. It is known that weights of reproductive organs are crucial benchmarks for the toxicological studies (Crissman *et al.*, 2004; Yavasoglu *et al.*, 2008). In our study, a significant decrease in the weight of testis was observed in lead-treated rats. It is considered crucial that the testis weight is largely dependent on the mass of differentiated spermatogenic cells and the reduction in the weight of testis might be due to disruption of spermatogenesis (Mruk & Cheng, 2004). The decrease in the testis weight observed in the present study is in consonance with earlier reports (Smith *et al.*, 2008; Wang *et al.*, 2008; Hamadouche *et al.*, 2009; Sainath *et al.*, 2011). The reduction in testosterone levels that was found in lead-treated rats is also one of the indicators of the chemical toxicity on reproductive system (Yoshida *et al.*, 2002). It has been suggested that the adequate bioavailability of testosterone plays an important role in the structural and functional integrity of reproductive organs (Mann, 1974) and the decrease in weights of these organs could be due to inadequate circulating male hormone. In addition, it is known that testosterone is essential to maintain the structure and function of the male accessory sex glands. Accessory glands such as prostate and seminal vesicle require androgen for differentiation, development and maintenance of epithelial cells (Ono *et al.*, 2004). Therefore, it can be interpreted that decrease in weights of prostate and seminal vesicle of lead-treated rats may be related to reduced levels of testosterone. One of the most sensitive tests for evaluating spermatogenesis is sperm count in the epididymis (Chandra *et al.*, 2007). Rats treated with lead acetate showed a reduction in cauda epididymis sperm concentration and motility and also the increase in the per cent number of morphologically abnormal spermatozoa. It has been reported that sperm counts may decrease as a result of reduction in testosterone levels (Uzunhisarcikli *et al.*, 2007). Moreover, these changes in epididymal sperm concentration are consistent with changes in epididymal weight in male rats.

Additionally, in the present study, lead toxicity caused testicular oxidative stress by increasing the levels of lipid peroxidation and decreasing the activities of SOD and

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GPx in testis. Our findings are compatible with the reports of Berry *et al.* (2002), Jensen *et al.* (2006) and Sainath *et al.* (2011). Lead-induced oxidative stress has been identified as the primary contributory agent in the pathogenesis of lead poisoning. ROS were generated as results of lead exposure have been identified in testis and spermatozoa (Hsu & Guo, 2002). Poor sperm quality caused by oxidative stress due to generation of ROS has been reported to result in infertility (Aitken & Baker, 2006). Generation of free radicals, subsequently attacks almost all cell components including lipid membrane, is known to play a crucial role in the aetiology of defective sperm function via elevation of lipid peroxidation in the plasma membrane (Agarwal & Saleh, 2002; Vernet *et al.*, 2004; Marchlewicz *et al.*, 2007; Ademuyiwa *et al.*, 2009). Moreover, lead binds to the sulphydryl groups or metal cofactors in antioxidant enzymes such as SOD and GPx, which reduce the antioxidant enzyme activities (Sivaprasad *et al.*, 2004; Patrick, 2006). Decrease in SOD and GPx activities leads to the oxidative stress in the tissue (Antonio-Garcia & Mass -Gonzalez, 2008). Our results showed that *F. parviflora* decreases the levels of lipid peroxidation and increases the antioxidant enzyme activities. The protective effects of *F. parviflora* may be due to the improvement in testicular oxidative status by its extract components. Haq & Hussain (1993) suggested that *Fumaria* species has ability to counteract oxidative stress with its antioxidative properties. Gilani *et al.* (1996) reported that the hepatoprotective activity of *F. parviflora* extract against paracetamol has been due to inhibitory effect on microsomal drug metabolising enzymes. Antioxidant components such as isoquinoline alkaloids, phenolic compounds and flavonoids have been reported in *Fumaria* species (Soušek *et al.*, 1999; Orhan *et al.*, 2010; Naz *et al.*, 2013). Inhibition of arthrosclerosis development by *F. vaillantii* hydroalcoholic extract in rabbit has been related to antioxidant effects of its flavonoids like rutin (Madani *et al.*, 2007). Phytochemical analyses have been shown that *Fumaria* species contains high amounts of isoquinoline alkaloids including fumaric acid (Jowkar *et al.*, 2011). It has been determined that fumaric acid content of *F. parviflora* to be about 0.93% w/w (Khalighi-Sigaroodi *et al.*, 2005). So, isoquinoline alkaloids are found to be the major contributors to the antioxidative activity in *F. parviflora*. Rao & Mishra (1998) suggested that fumaric acid is probably responsible for the antioxidative activity of *F. indica*. It has been shown that fumarates may activate the Nrf2 transcriptional pathway, which is known to mediate induction of phase 2 genes by SH-reactive electrophiles and to play a major role in cell and tissue defence against oxidative stress (Jung & Kwalk, 2010). A result of the activated Nrf2 pathway is cellular detoxification, normalisation of the energy metabolism and repair and/or degradation of damaged proteins (Ellrichmann *et al.*, 2011). It also reported that fumarate potently increases the expression of antioxidant response element (ARE)-regulated genes (e.g. Gsta1, Nqo1 and Hmox1) (Kwak *et al.*, 2003; Hayes *et al.*, 2010) through activation of the redox-sensitive transcription factor Nrf2 (Hayes & McMahon, 2009; Taguchi *et al.*, 2011). In addition, Ashrafian *et al.* (2012) proposed that clinically established fumarate derivatives activate the Nrf2 pathway and are readily testable cytoprotective agents.

Conclusions

It could be concluded that co-administration of ethanolic extract of *F. parviflora* to lead acetate-treated rats ameliorates the lead-induced oxidative stress by decreasing lipid peroxidation and activating antioxidant enzymes in the testis, thereby significantly improves the reproductive parameters in male rats. This study suggests the protective effect of *F. parviflora* to minimise the hazardous effects of lead exposure. Further studies are needed to evaluate the therapeutic efficacy and toxicity of isolated compounds of *F. parviflora* in various animal models.

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